AIVOA PHARMA SCIENCES LTD. - API MANUFACTURING DIVISION

GMP DEVIATION & PRELIMINARY INVESTIGATION REPORT (ICH Q7)

Deviation Reference ID: DEV-API-2026-089
Date of Incident: 2026-10-24 14:20 HRS
API Product Name: Atorvastatin Calcium API (USP/EP Grade)
Manufacturing Stage: Stage-III Final Crystallization & Isolation
Batch Number: ATV-261004 (Batch Size: 450.0 kg)
Equipment ID: Glass-Lined Reactor GLR-204 & Chilled Brine Loop CBL-02
Manufacturing Site / Block: API Synthesis Block-B, Cleanroom Suite 4
Preliminary Classification: CRITICAL DEVIATION

1. INCIDENT DESCRIPTION & PROCESS EXCURSION:

During Stage-III crystallization of Atorvastatin Calcium Batch ATV-261004 in Glass-Lined Reactor GLR-204, the BMR step requires linear cooling of the reaction mass from 65.0 C to 15.0 C over 300 minutes (target cooling rate: 0.16 C/min) to ensure selective crystallization of Polymorph Form I. At 14:20 HRS, SCADA alarm ALM-204-T triggered. The reactor jacket cooling stalled at 43.5 C for 95 minutes before operator intervention opened the manual bypass valve.

2. CRITICAL QUALITY ATTRIBUTES (CQA) & QC LAB OOS RESULTS:

- In-Process HPLC Purity Check (Sample ID: QC-ATV-882):
  * Desfluoro Impurity: 0.08% (Spec: NMT 0.15%) - PASS
  * Diastereomer Impurity (Impurity-D): 0.27% (Spec: NMT 0.15%) - FAIL (OOS)
- Powder X-Ray Diffraction (PXRD): Detected ~18% unwanted amorphous/Form-III phase.

3. IMMEDIATE CONTAINMENT ACTIONS TAKEN:

- Batch ATV-261004 (wet cake, 462 kg) immediately placed under QA QUARANTINE hold tag #QH-904.
- Reactor GLR-204 tagged 'OUT OF SERVICE - ENGINEERING HOLD' pending valve inspection.
- Notification sent to Head of Quality Assurance and Qualified Person (QP).

4. INITIAL OBSERVATIONS & AVAILABLE EVIDENCE:

- SCADA trend log confirms PLC sent 100% open command signal (20 mA) to chilled brine proportional control valve CV-204B, but flow meter FM-204 showed brine flow dropped to 1.2 m3/hr (normal: 14 m3/hr).
- Operator shift handover log noted slight air hissing near CV-204B pneumatic actuator 2 days prior, but no maintenance work order was raised.
- Last Preventive Maintenance (PM) on control valve CV-204B was performed 14 months ago (SOP requires annual diaphragm and positioner PM every 6 months for critical crystallization reactors).

5. OPEN EVIDENCE GAPS TO VERIFY ROOT CAUSE:

- Physical dismantling and bench pressure test report of pneumatic actuator diaphragm on CV-204B.
- Calibration verification certificate of reactor RTD temperature probe TT-204A.
- Purity & water content CoA verification of incoming Methanol solvent Lot #METH-9921.